

Linear Algebra Primer

1. Anton, Howard, “Elementary Linear Algebra,” 1977, *John Wiley&Sons*
2. Cullen, Charles G., “Matrices and Linear Transformations,” 1972, 2nd Ed., *Addison-Wesley*,
Reprinted by Dover 1990.
3. Strang, Gilbert, “Linear Algebra and Its Applications,” 1988, 3rd Ed., *International Thomson Publishing.*

Scalars, Vectors, Matrices and Tensors . .	
Multiplying Matrices and Vectors	
Identity and Inverse Matrices	
Linear Dependence and Span	
Norms	
Special Kinds of Matrices and Vectors . .	
Eigendecomposition	
Singular Value Decomposition	
The Moore-Penrose Pseudoinverse	
The Trace Operator	
The Determinant	
Example: Principal Components Analysis	

- Vectors:

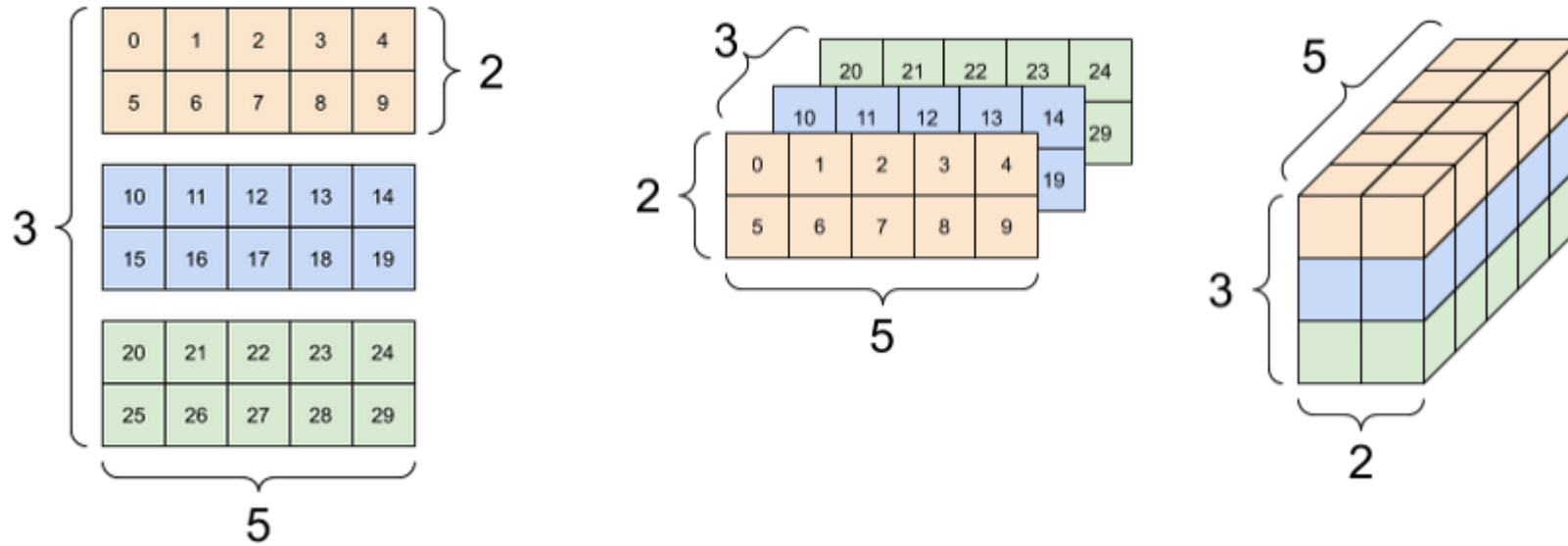
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

- Matrices: A matrix is a 2-D array of numbers, $\mathbf{A} \in \mathbb{R}^{m \times n}$

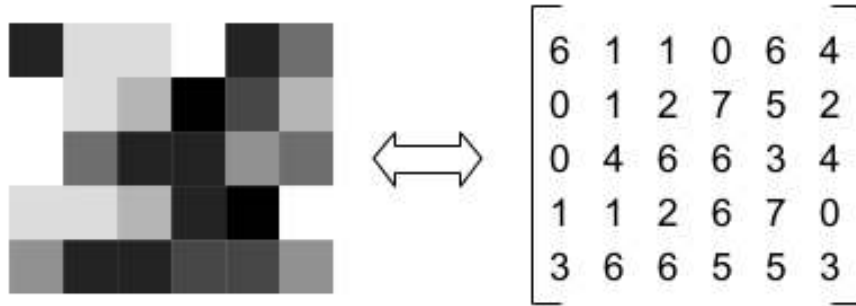
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1k} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2k} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3k} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{j1} & a_{j2} & a_{j3} & \cdots & a_{jk} & \cdots & a_{jn} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mk} & \cdots & a_{mn} \end{bmatrix}$$

- **Tensors:** In some cases we will need an array with more than two axes. In the general case, an array of numbers arranged on a regular grid with a variable number of axes is known as a tensor. We denote a tensor named “A” with this typeface: **A**. We identify the element of **A** at coordinates (i, j, k) by writing $A_{i,j,k}$.

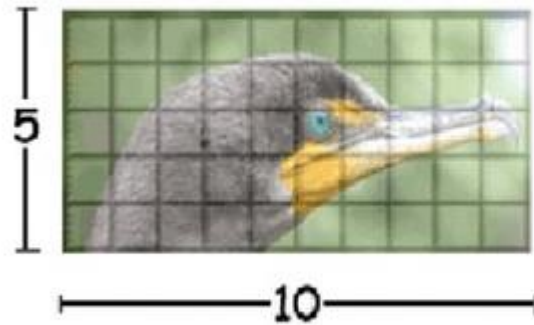
A 3-axis tensor, shape: [3, 2, 5]



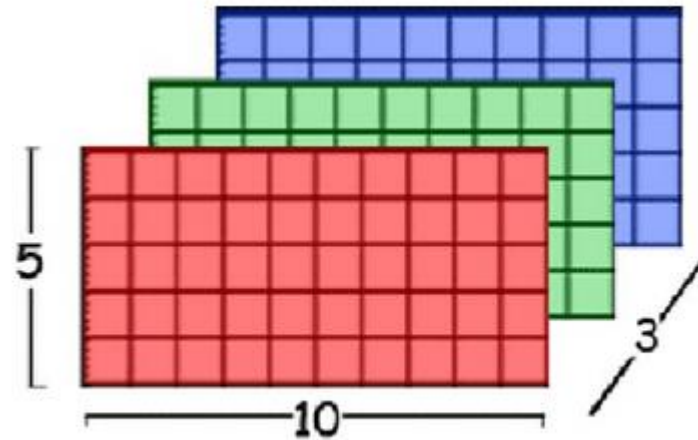
<https://www.tensorflow.org/guide/tensor>



<https://stackoverflow.com/questions/16354864/how-are-non-integer-images-represented>

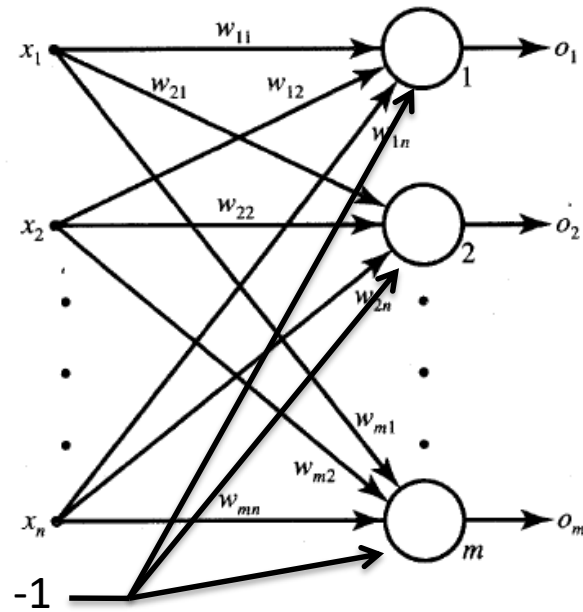


Original Color Image



Matlab RGB Matrix

<https://dev.to/tbhaxor/pytorch-understanding-tensors-part-2-4bfc>



- mathematical equations?

componentwise equations

$$v_1 = w_{11}x_1 + w_{12}x_2 + \dots + w_{1n}x_n - \theta_1 \quad o_1 = f(v_1)$$

⋮

$$v_i = w_{i1}x_1 + w_{i2}x_2 + \dots + w_{in}x_n - \theta_i \quad o_i = f(v_i)$$

⋮

$$v_m = w_{m1}x_1 + w_{m2}x_2 + \dots + w_{mn}x_n - \theta_m \quad o_m = f(v_m)$$

Linear Sum

Output

vectorial notation

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad w_i = \begin{pmatrix} w_{i1} \\ w_{i2} \\ \vdots \\ w_{in} \end{pmatrix} \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix} \quad o = \begin{pmatrix} o_1 \\ o_2 \\ \vdots \\ o_m \end{pmatrix}$$

input

i th neuron weight

linear sum

output

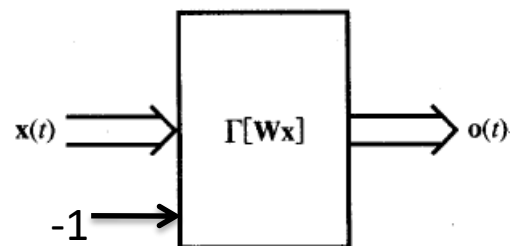
$$v_i = w_i^T x - \theta_i \quad o_i = f(v_i) \quad i = 1, 2, \dots, m$$



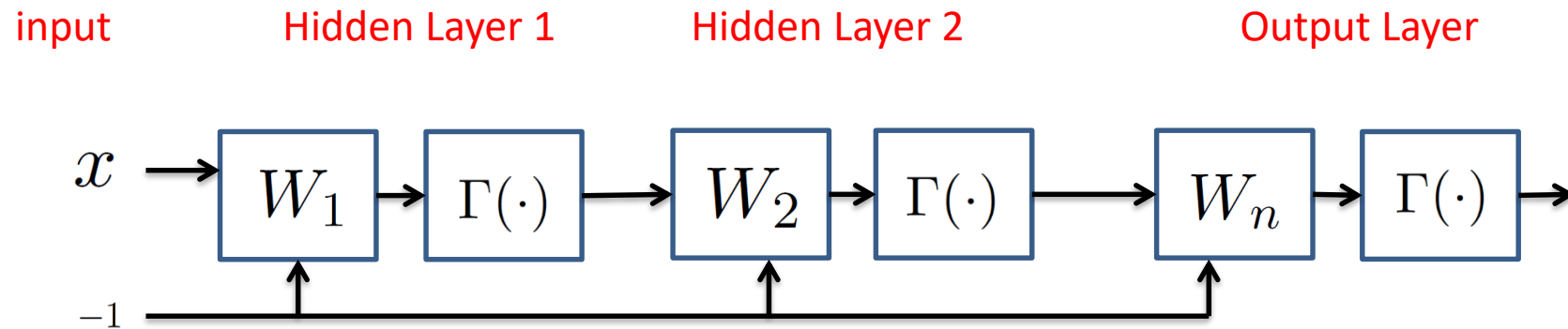
number of neurons

$$W = \begin{pmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_m^T \end{pmatrix} = \begin{pmatrix} w_{11} & w_{12} & \dots & w_{1n} \\ w_{21} & w_{22} & \dots & w_{2n} \\ w_{m1} & w_{m2} & \dots & w_{mn} \end{pmatrix} \quad \theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_m \end{pmatrix}$$

$$\Gamma(v) = \begin{pmatrix} f(v_1) & 0 & \dots & 0 \\ 0 & f(v_2) & \dots & 0 \\ & & \vdots & \\ 0 & 0 & \dots & f(v_m) \end{pmatrix}$$



$$v = Wx - \theta \quad o = \Gamma(v) = \Gamma(Wx - \theta)$$



$$o = \Gamma(W_n(\Gamma(\dots W_2\Gamma(W_1x - \theta_1) - \theta_2)\dots) - \theta_n)$$

Could also use extended notation

The transpose of a matrix $\mathbf{A}$, written as $\mathbf{A}^T$,

$$(\mathbf{A}^T)_{i,j} = A_{j,i}.$$

1. $(\mathbf{A}^T)^T = \mathbf{A}$
2. $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
3. $(k\mathbf{A})^T = k\mathbf{A}^T$
4. $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$

We can add matrices to each other, as long as they have the same shape.

$$\mathbf{C} = \mathbf{A} + \mathbf{B} \quad \longleftrightarrow \quad C_{i,j} = A_{i,j} + B_{i,j}.$$

One of the most important operations involving matrices is multiplication of two matrices. The **matrix product** of matrices $\mathbf{A}$ and $\mathbf{B}$ is a third matrix $\mathbf{C}$. In order for this product to be defined, $\mathbf{A}$ must have the same number of columns as $\mathbf{B}$ has rows. If $\mathbf{A}$ is of shape $m \times n$ and $\mathbf{B}$ is of shape $n \times p$, then $\mathbf{C}$ is of shape $m \times p$. We can write the matrix product just by placing two or more matrices together, e.g.

$$\mathbf{C} = \mathbf{AB} \quad \longleftrightarrow \quad C_{i,j} = \sum_k A_{i,k} B_{k,j}.$$

Note that the standard product of two matrices is *not* just a matrix containing the product of the individual elements. Such an operation exists and is called the **element-wise product** or **Hadamard product**, and is denoted as $\mathbf{A} \odot \mathbf{B}$.

The **dot product** between two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ of the same dimensionality is the matrix product $\boldsymbol{x}^\top \boldsymbol{y}$. We can think of the matrix product $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ as computing $C_{i,j}$ as the dot product between row i of $\boldsymbol{A}$ and column j of $\boldsymbol{B}$.

$$\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$$

$$\boldsymbol{A} + (\boldsymbol{B} + \boldsymbol{C}) = (\boldsymbol{A} + \boldsymbol{B}) + \boldsymbol{C}$$

$$\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{A}\boldsymbol{B} + \boldsymbol{A}\boldsymbol{C}$$

$$\boldsymbol{A}(\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}.$$

Note that in general, matrices are not commutative over multiplication:

$$\boldsymbol{A}\boldsymbol{B} \neq \boldsymbol{B}\boldsymbol{A}$$

However, the dot product between two vectors

$$\boldsymbol{x}^\top \boldsymbol{y} = \boldsymbol{y}^\top \boldsymbol{x}.$$

$$\mathbf{AB} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

$$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix}$$

Linear Systems of Equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{cases}$$

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a known matrix, $\mathbf{b} \in \mathbb{R}^m$ is a known vector, and $\mathbf{x} \in \mathbb{R}^n$ is a vector of unknown variables we would like to solve for.

Theorem 1.1. *Any system of linear equations has one of the following exclusive conclusions.*

- (a) *No solution.*
- (b) *Unique solution.*
- (c) *Infinitely many solutions.*

A linear system is said to be **consistent** if it has at least one solution; and is said to be **inconsistent** if it has no solution.

Geometric interpretation

The following three linear systems

$$(a) \begin{cases} 2x_1 + x_2 = 3 \\ 2x_1 - x_2 = 0 \\ x_1 - 2x_2 = 4 \end{cases} \quad (b) \begin{cases} 2x_1 + x_2 = 3 \\ 2x_1 - x_2 = 5 \\ x_1 - 2x_2 = 4 \end{cases} \quad (c) \begin{cases} 2x_1 + x_2 = 3 \\ 4x_1 + 2x_2 = 6 \\ 6x_1 + 3x_2 = 9 \end{cases}$$

have no solution, a unique solution, and infinitely many solutions, respectively. See Figure 1.

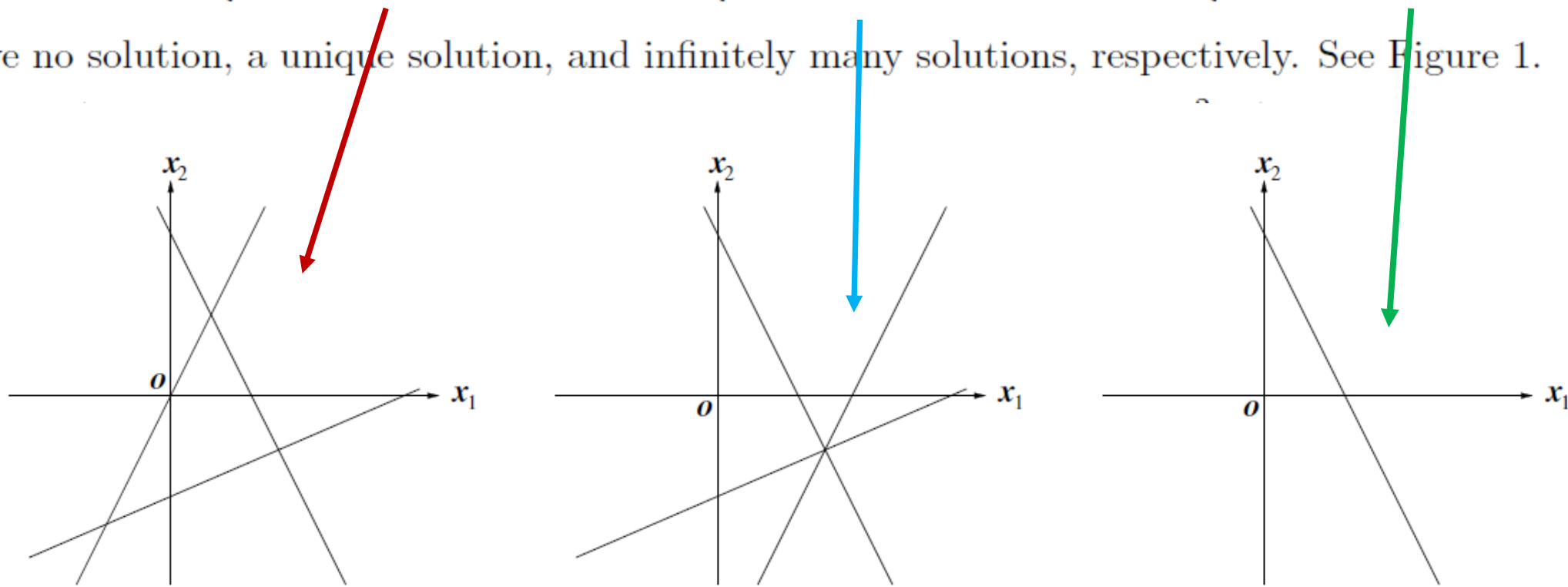


Figure 1: No solution, unique solution, and infinitely many solutions.

$$\mathbf{A}\mathbf{x} = \sum_i x_i \mathbf{A}_{:,i}.$$

In general, this kind of operation is called a **linear combination**. Formally, a linear combination of some set of vectors $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ is given by multiplying each vector $\mathbf{v}^{(i)}$ by a corresponding scalar coefficient and adding the results:

$$\sum_i c_i \mathbf{v}^{(i)}.$$

The **span** of a set of vectors is the set of all points obtainable by linear combination of the original vectors.

The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of the columns of A .

Given the linear system $A\mathbf{x} = B$ and the augmented matrix $(A|B)$.

- ① If $\text{rank}(A) = \text{rank}(A|B) =$ the number of rows in x , then the system has a unique solution.
- ② If $\text{rank}(A) = \text{rank}(A|B) <$ the number of rows in x , then the system has ∞ -many solutions.
- ③ If $\text{rank}(A) < \text{rank}(A|B)$, then the system is inconsistent.

Identity and Inverse Matrices

$$A^{-1}A = I \quad AA^{-1} = I \quad \text{if } A^{-1} \text{ exists,}$$

$$I = \begin{bmatrix} 1 & 0 & 0 & . & . \\ 0 & 1 & 0 & . & . \\ 0 & 0 & 1 & . & . \\ . & . & . & . & . \\ . & . & . & . & 1 \end{bmatrix}$$

$$Ax = b \quad \Rightarrow \quad x = A^{-1}b$$

Determinant

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & -2 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$\det A = 2(-2) - 1(3) = -7$$

$$\det A = 2(-2(2) - (-1)(1)) - 1(3(2) - 1(1))$$

$$+ 0(3(-1) - (-2)(1)) = 2(-3) - 5 = -11$$

2 multiplications

9 multiplications

Laplace expansion

 21 languages 

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This article is about the expression of a determinant in terms of minors. For the approximation of radial potentials, see [Laplace expansion \(potential\)](#).

In [linear algebra](#), the **Laplace expansion**, named after [Pierre-Simon Laplace](#), also called **cofactor expansion**, is an expression of the [determinant](#) of an $n \times n$ [matrix](#) B as a weighted sum of [minors](#), which are the determinants of some $(n - 1) \times (n - 1)$ [submatrices](#) of B . Specifically, for every i ,

$$\det(B) = \sum_{j=1}^n (-1)^{i+j} b_{i,j} m_{i,j},$$

https://en.wikipedia.org/wiki/Laplace_expansion

Leibniz formula for determinants

 8 languages 

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In [algebra](#), the **Leibniz formula**, named in honor of [Gottfried Leibniz](#), expresses the [determinant](#) of a [square matrix](#) in terms of permutations of the matrix elements. If A is an $n \times n$ matrix, where a_{ij} is the entry in the i -th row and j -th column of A , the formula is

$$\det(A) = \sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_{i=1}^n a_{i, \tau(i)} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{\sigma(i), i}$$

https://en.wikipedia.org/wiki/Leibniz_formula_for_determinants

Computational complexity

https://en.wikipedia.org/wiki/Computational_complexity_of_mathematical_operations

3.5 Properties of Determinant Operations

1. If $\mathbf{A}$ is a square matrix then $\det(\mathbf{A}^T) = \det(\mathbf{A})$
2. $\det(k\mathbf{A}) = k^n \det(\mathbf{A})$ where $\mathbf{A}$ is an $n \times n$ matrix and k is a scalar
3. $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ where $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size.
4. A square matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$
5. For a homogeneous system of linear equations $\mathbf{Ax} = \mathbf{0}$

There exists a nontrivial $\mathbf{x}$ ($\mathbf{x} \neq \mathbf{0}$) if and only if $\det(\mathbf{A}) = 0$

6. If $\mathbf{A}$ is invertible then

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A}) \text{ where } \text{adj} \mathbf{A} \text{ is the adjoint of } \mathbf{A}.$$

The adjoint of a matrix $\mathbf{A}$ is defined as the transpose of the cofactor matrix

Cramer's rule

https://en.wikipedia.org/wiki/Cramer%27s_rule

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$



$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})} \quad i = 1, \dots, n$$

where $\mathbf{A}_i$ is the matrix formed by replacing the i -th column of $\mathbf{A}$ by the column vector $\mathbf{b}$.

$$\begin{bmatrix} 2 & 4 & -2 \\ 0 & 2 & 3 \\ 1 & 0 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 18 \\ -2 \\ -7 \end{pmatrix} \quad \longrightarrow \quad \mathbf{A}_1 = \begin{bmatrix} 18 & 4 & -2 \\ -2 & 2 & 3 \\ -7 & 0 & 5 \end{bmatrix} \quad \mathbf{A}_2 = \begin{bmatrix} 2 & 18 & -2 \\ 0 & -2 & 3 \\ 1 & -7 & 5 \end{bmatrix} \quad \mathbf{A}_3 = \begin{bmatrix} 2 & 4 & 18 \\ 0 & 2 & -2 \\ 1 & 0 & -7 \end{bmatrix}$$

$$x_1 = \frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})} = \frac{108}{36} = 3, \quad x_2 = \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})} = \frac{72}{36} = 2, \quad x_3 = \frac{\det(\mathbf{A}_3)}{\det(\mathbf{A})} = -\frac{72}{36} = -2$$

$$A^{-1} = ?$$

$$P [A \ I] = [I \ X]$$

$$X = A^{-1}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 2 & 3 & -14 & 6 & 3 \\ 0 & 1 & -3 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

- Norms in V :

A function $\| \cdot \| : V \rightarrow \mathbf{R}_+$ is called a **norm** if

- **1** : $\|x\| = 0 \iff x = 0$
- **2** : $\|\alpha x\| = |\alpha| \|x\| \quad \forall x \in V, \forall \alpha$
- **3** : $\|x + y\| \leq \|x\| + \|y\| \quad \forall x, y \in V$ (triangle inequality)

- Practically, $\|x\|$ gives a measure of the **length** of x

- **Typical Example :** L_p norm

- Let $V = \mathbf{R}^n$ and $1 \leq p < \infty$

- $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} = \left(|x_1|^p + \dots + |x_n|^p \right)^{1/p}$

Minkowski's inequality :

- $(\sum_i^n (a_i + b_i)^p)^{1/p} \leq (\sum_i^n (a_i)^p)^{1/p} + (\sum_i^n (b_i)^p)^{1/p} \quad p \geq 1, a_i, b_i \geq 0$

- For $p = 1$, called 1-norm
- $\|x\|_1 = \sum_{i=1}^n |x_i| = |x_1| + \dots + |x_n|$

- For $p = 2$, standard Euclidean norm.
- $\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{x^T x} \quad \Rightarrow \quad \|x\|_2^2 = x^T x$

- For $p = \infty$, called ∞ -norm
- $\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$

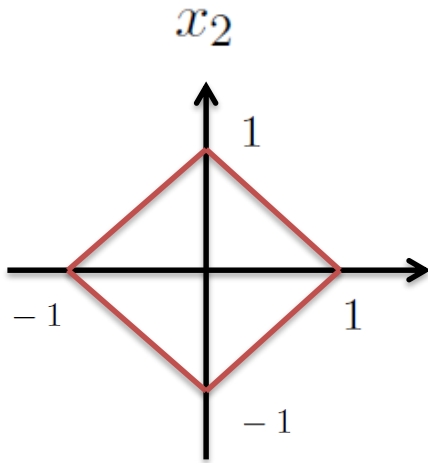
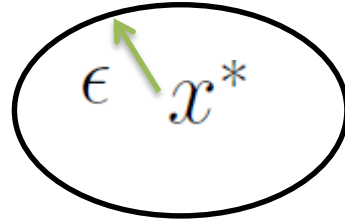
$$x = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \begin{array}{l} \nearrow \|x\|_1 = 7 \\ \longrightarrow \|x\|_2 = 5 \\ \searrow \|x\|_\infty = 4 \end{array}$$

- For $p = 1$, called 1-norm
- $\|x\|_1 = \sum_{i=1}^n |x_i| = |x_1| + \dots + |x_n|$

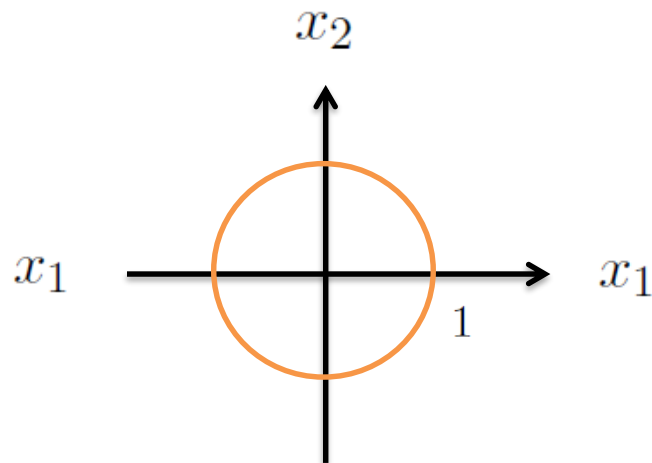
- For $p = 2$, standard Euclidean norm.
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- For $p = \infty$, called ∞ -norm
- $\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$

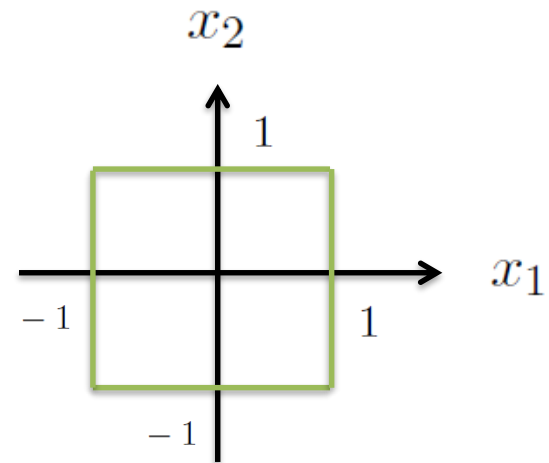
$$x = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \begin{array}{l} \nearrow \|x\|_1 = 7 \\ \longrightarrow \|x\|_2 = 5 \\ \searrow \|x\|_\infty = 4 \end{array}$$



$$\|x\|_1 = |x_1| + |x_2| = 1$$



$$\|x\|_2 = \sqrt{x_1^2 + x_2^2} = 1$$



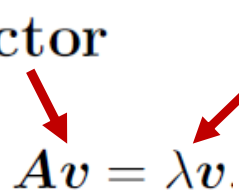
$$\|x\|_\infty = \max\{|x_1|, |x_2|\} = 1$$

Sometimes we may also wish to measure the size of a matrix. In the context of deep learning, the most common way to do this is with the otherwise obscure **Frobenius norm**:

$$||A||_F = \sqrt{\sum_{i,j} A_{i,j}^2},$$

$$\boldsymbol{x}^\top \boldsymbol{y} = ||\boldsymbol{x}||_2 ||\boldsymbol{y}||_2 \cos \theta,$$

eigenvector eigenvalue


$$Av = \lambda v.$$

$$(\lambda I - A)v = 0 \quad \longleftrightarrow \quad \det(\lambda I - A) = 0 \quad \Longrightarrow \quad \lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} \dots + c_n = 0$$

$$\mathbf{A} = \begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix}$$

$$\det \begin{bmatrix} 4 - \lambda & -5 \\ 1 & -2 - \lambda \end{bmatrix} = \lambda^2 - 2\lambda - 3 = 0 \quad \begin{array}{l} \lambda_1 = -1 \\ \lambda_2 = 3 \end{array}$$

$$(\lambda_1 I - A) x^1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

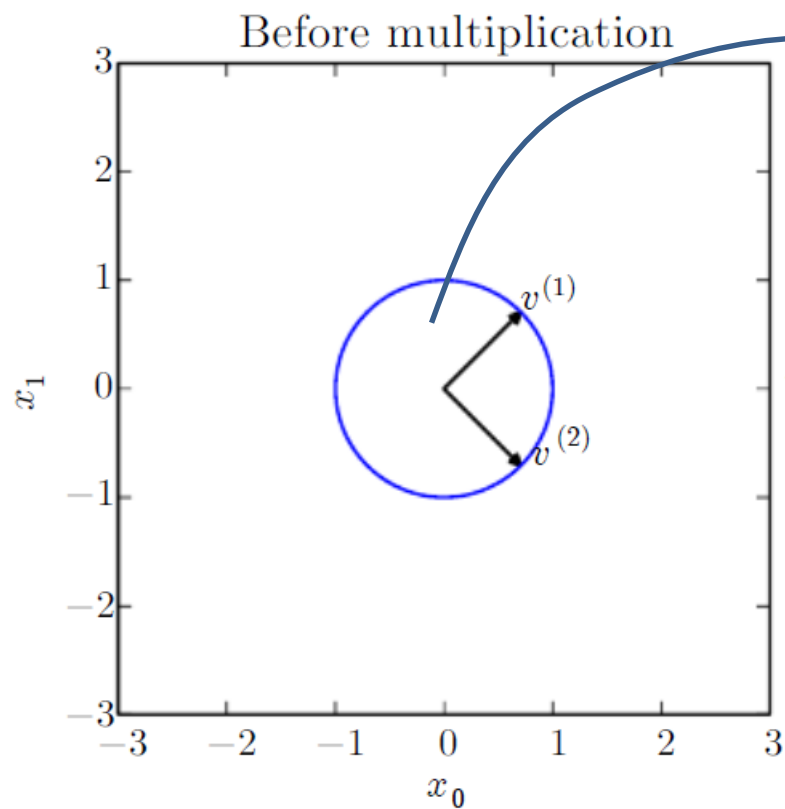
$$(\lambda_2 I - A) x^2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\mathbf{x}^1 = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

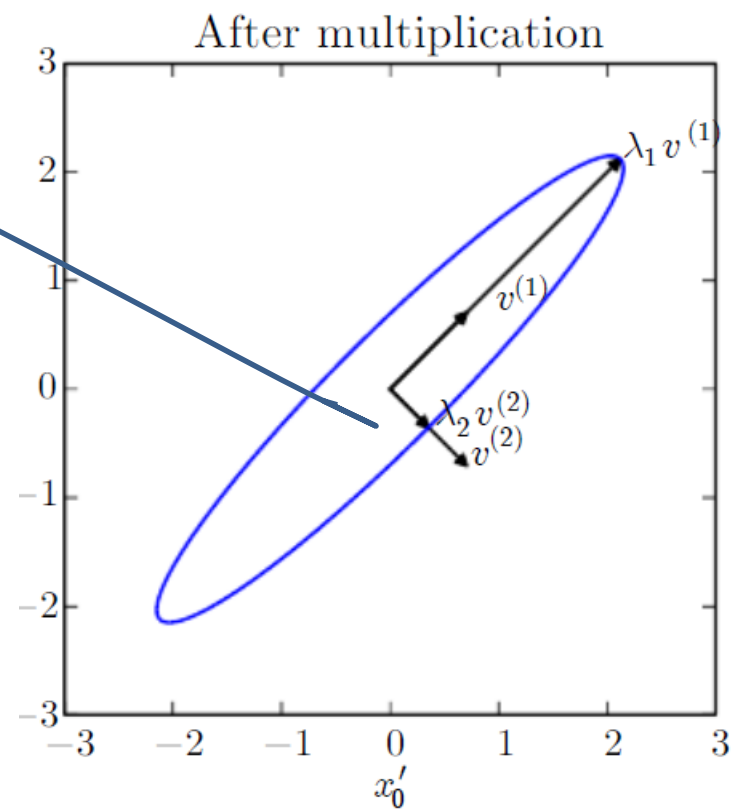
$$\mathbf{x}^2 = c_2 \begin{pmatrix} 5 \\ 1 \end{pmatrix}$$

$$A \begin{bmatrix} x^1 & x^2 \end{bmatrix} = \begin{bmatrix} x^1 & x^2 \end{bmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \quad \longrightarrow \quad A = \begin{bmatrix} x^1 & x^2 \end{bmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \begin{bmatrix} x^1 & x^2 \end{bmatrix}^{-1}$$

$$\mathbf{A} = \mathbf{V} \text{diag}(\boldsymbol{\lambda}) \mathbf{V}^{-1}$$



$A :$



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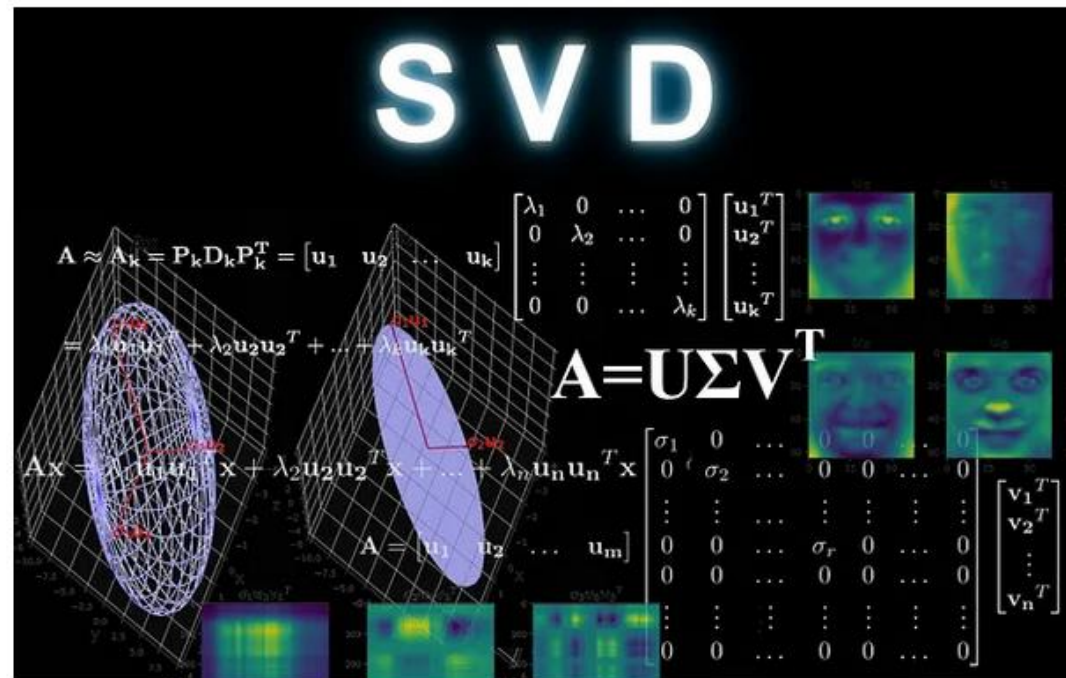
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https://en.wikipedia.org/wiki/Singular_value_decomposition

every $m \times n$ matrix A can be factored as

$$A = U\Sigma V^T$$

- U is $m \times m$ and orthogonal $\Rightarrow UU^T = U^T U = I_m$
- V is $n \times n$ and orthogonal $\Rightarrow VV^T = V^T V = I_n$
- Σ is $m \times n$ and “diagonal”: diagonal with diagonal elements $\sigma_1, \dots, \sigma_n$ if $m = n$,

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \quad \text{if } m > n, \quad \Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \sigma_m & 0 & \cdots & 0 \end{bmatrix} \quad \text{if } m < n$$

- the diagonal entries of Σ are nonnegative and sorted:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min\{m,n\}} \geq 0$$

$$AV = U\Sigma, \quad A^T U = V\Sigma^T$$

$$Av_i = \sigma_i u_i \quad \text{and} \quad A^T u_i = \sigma_i v_i \quad \text{for } i = 1, \dots, \min\{m, n\}$$

$$U \sim \text{eigenvectors of } AA^T$$

$$V \sim \text{eigenvectors of } A^T A$$

$$\Sigma \sim \sqrt{\text{eigenvalues}(A^T A)} = \sqrt{\text{eigenvalues}(AA^T)}$$

Example Problem : $Ax = b$ when A^{-1} does not exist

Least Squares Solution $\min_x \|Ax - b\|_2$

(Moore-Penrose) Pseudo-inverse $x^* = A^+ b$

https://en.wikipedia.org/wiki/Moore%E2%80%93Penrose_inverse

$$\text{If } \mathbf{A} \text{ has full column rank} \quad (\text{rank}(\mathbf{A}) = n \leq m) \quad \mathbf{A}^+ = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$$

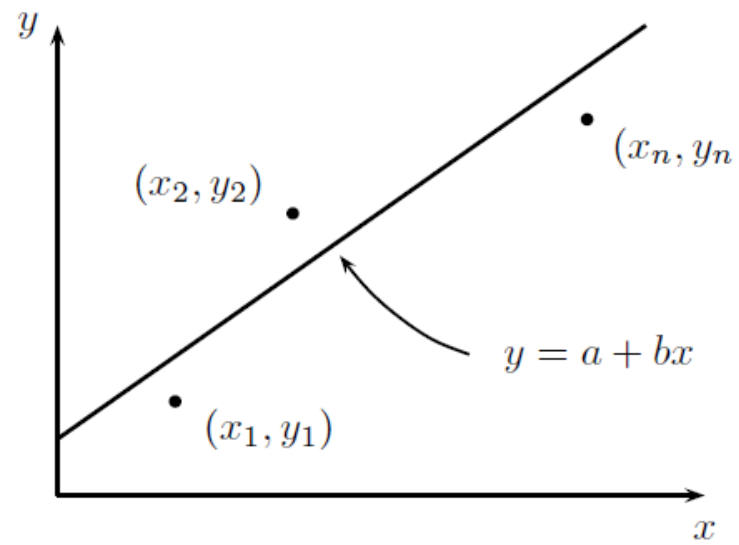
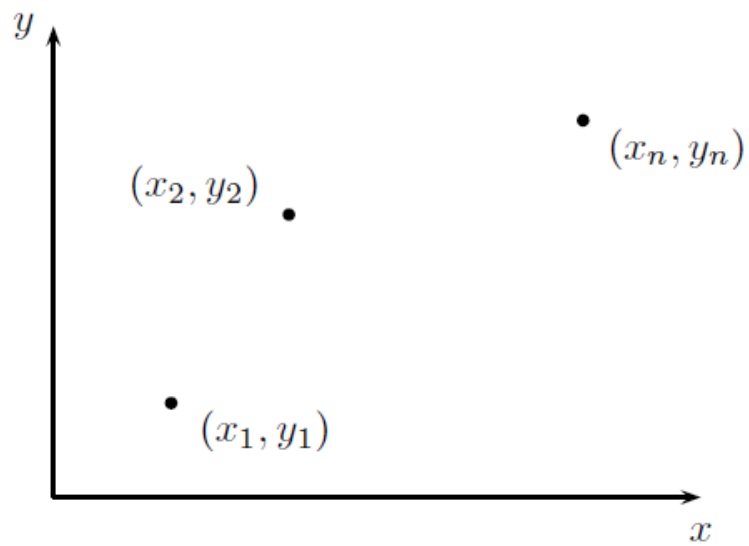
$$\text{If } \mathbf{A} \text{ has full row rank} \quad (\text{rank}(\mathbf{A}) = m \leq n) \quad \mathbf{A}^+ = \mathbf{A}^T (\mathbf{A} \mathbf{A}^T)^{-1}$$

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \implies \mathbf{A}^+ = \mathbf{V} \mathbf{\Sigma}^+ \mathbf{U}^T$$

$$\mathbf{A}^+ = \lim_{\alpha \searrow 0} (\mathbf{A}^\top \mathbf{A} + \alpha \mathbf{I})^{-1} \mathbf{A}^\top.$$

$$\begin{array}{rcl} 1.3x_1 + 0.6x_2 & = & 3.3 \\ 4.7x_1 + 1.5x_2 & = & 13.5 \\ 3.1x_1 + 5.2x_2 & = & -0.1 \end{array} \quad \mathbf{A} = \begin{bmatrix} 1.3 & 0.6 \\ 4.7 & 1.5 \\ 3.1 & 5.2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3.3 \\ 13.5 \\ -0.1 \end{bmatrix}$$

$$\bar{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} = \begin{bmatrix} 3.5526 \\ -2.1374 \end{bmatrix}$$



$$\begin{aligned} a + x_1 b &= y_1 \\ a + x_2 b &= y_2 \\ &\vdots \\ a + x_n b &= y_n \end{aligned}$$

 $\Leftrightarrow$

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \\ 1 & x_n \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$\begin{matrix} \text{A} \uparrow & & \downarrow \end{matrix}$

$$(A^T A)^{-1} A^T \mathbf{b}.$$

$A\mathbf{x} = \mathbf{b}$ has no solution means that $\mathbf{b}$ is not in the column space of A . The least squares solution to $A\mathbf{x} = \mathbf{b}$ is simply the vector $\bar{\mathbf{x}}$ for which $A\bar{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto the column space of A . This is shown simplistically below, for the situation where the column space is a plane in $\mathbb{R}^3$.

